Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech ***Time: 03 Hours***
Code: 101605 ***Subject: Environmental Engineering - II*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** [2 x 7 = 14]

(a) Quantity of sewage mainly depends on:
(i) Temperature
(ii) Soil type
(iii) Water supply rate
(iv) pH of water.

(b) Peak factor in sewer design is used to account for:
(i) Slope
(ii) Variation in sewage flow
(iii) Sewer blockages
(iv) Structural load

(c) Activated sludge process is an example of:
(i) Anaerobic attached growth
(ii) Aerobic suspended growth
(iii) Anaerobic suspended growth
(iv) Aerobic attached growth

(d) Trickling filter is an example of:
(i) Aerobic attached growth
(ii) Anaerobic attached growth
(iii) Anaerobic suspended growth
(iv) Chemical treatment

(e) Municipal Solid Waste (MSW) mainly includes:
(i) Hazardous chemicals
(ii) Household waste
(iii) Nuclear waste
(iv) Agricultural waste only

(f) One of the most effective forms of hazardous waste disposal may be
(i) land fills
(ii) deep well injection
(iii) Incineration
(iv) Surface impoundments

(g) The bacteria which require free oxygen for their survival are called
(i) aerobic bacteria
(ii) anaerobic bacteria
(iii) facultative bacteria
(iv) None of these

(h) Biomedical Waste Rules are notified by:
(i) MoEF&CC
(ii) Home Ministry
(iii) Ministry of Health
(iv) NITI Aayog

(i) Incineration is suitable for waste with:
(i) High calorific value
(ii) High moisture content
(iii) High metal content
(iv) Inert material

(j) BOD stands for:
(i) Basic oxygen demand
(ii) Biochemical oxygen demand
(iii) Biological oxidation demand
(iv) Bio-organic decomposition

Q.2 (a) Explain how central and state agencies coordinate in managing solid waste, sewerage, and pollution control in India. [7]

(b) Explain the environmental impacts of biomedical waste. Describe its segregation, collection, treatment, and disposal as per Biomedical Waste Management Rules. [7]

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Q.3 (a) Rectangular grit chamber is designed to remove particle with a diameter 0.2 mm and specific gravity 2.65. The settling velocities of these particles are found to be 0.02 m/s. A flow through velocity of 0.30 m/s will be maintained by the proportioning weir. Determine the channel dimensions for a maximum wastewater flow of 10,000 m/day. [7]

(b) Define hazardous waste. Explain the types and characteristics of hazardous wastes as per the Hazardous Waste Rules and CPCB schedules. [7]

Q.4 (a) Discuss in detail the various shapes of sewers used in sewerage systems. Explain their suitability, advantages, and limitations. [7]

(b) A city has a projected population of 60,000 spread over an area of 50 hectare. Assume rate of water supply of 250 litres per capita per day, and out of this total supply only 75 % reaches in sewer as wastewater. Assume a peak factor of 1.5. Design a separate circular sewer line running half full. Consider Manning's rugosity coefficient of n = 0.012, and gradient of sewer S = 0.0001. [7]

Q.5 (a) Explain the importance of the following in the design of sewers: [7]
(i) Self cleansing velocity and (ii) Non scouring velocity

(b) A 2 m deep trickling filter with a diameter of 20 m is operated with a recirculation ratio of 1.5. The raw wastewater flow rate is 3 ML/d and the 5-day BOD of the raw wastewater is 210 mg/L. Assuming that the primary tank BOD removal efficiency is 40%, compute the hydraulic filter. as well as organic load on the trickling [7]

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Q.6 (a) An oxidation pond is to be constructed in a hot climatic area for treating the sewage from a residential colony with 5000 persons contributing sewage at 120 litres/capita/day. The 5day BOD of sewage is 300mg/ litre. Assume organic loading in the pond as 300kg/ hectares/day. Calculate the dimensions required for the tank and its detention time. [7]

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(b) Explain the features of a sludge digestion tank with a neat sketch. [7]

Q.7 (a) Explain the following: [7]
(i) Pyrolysis
(ii) Open dumping
(iii)Refuse derived fuel
(iv) Biogasification

(b) Discuss in detail the zones of pollution in a river or stream. [7]

Q.8 (a) Design an imhoff tank to treat the sewage from a small town with a population of 20000 persons, with sewage flow rate of 180 litres per day. [7]

(b) What are the main challenges in MSW management in Indian cities? Provide engineering, administrative, and community-based solutions. [7]

Q.9 (a) A raw wastewater with the following characteristics is being treated in an activated sludge plant. Given that. BOD_5 of raw wastewater 220 gm/l; effluent BOD_5 desired = 20 mg/l; inflow rate to aeration tank = 100m^3/d; mixed liquor volatile suspended solid concentration 2200 mg/l; volume of aeration tank = 500m^3; water depth in aeration tank = 2.5 m. Determine the [7]
(i) food to microorganisms (F/M) ratio
(ii) space loading on aeration tank
(iii) hydraulic loading on aeration tank
(iv) detention period in aeration tank

(b) Distinguish between aerobic and anaerobic wastewater treatment processes. Explain suspended and attached growth systems with suitable examples. [7]

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